Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 35: Order Does Not Matter

Combinations

Chartrand & Zhang, Section 8.4

Name: _____ Date: _____

Where we are

Monday counted ordered selections. Most real problems are not ordered — a committee, a poker hand, a subset. Today's formula is Monday's divided by the number of orderings we do not want to distinguish.

By the end of today you should be able to

1. Compute $C(n, r)$ and explain the division by $r!$.
2. Decide whether a problem is a permutation or a combination.
3. Use $C(n, r) = C(n, n - r)$ and the subset-count identity.

1. Combinations

Definition

An **r-combination** of a set is an *unordered* selection of r of its elements — that is, an r-element **subset**. The number of them is written $C(n, r)$ or $\binom{n}{r}$.

Derive the formula by over-counting, then correcting. Each r-element subset can be arranged in $r!$ different orders, and each of those orders is a distinct r-permutation. So

$$P(n, r) = C(n, r) \cdot r! \qquad \Longrightarrow \qquad C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{r!\,(n - r)!}.$$

Check it. $P(8, 3) = 336$ and $3! = 6$, so $C(8, 3) = 56$.

Divide by what you over-counted

This is a technique, not just a formula. **Count with order, then divide by the number of orderings that produce the same outcome.** It reappears whenever a count distinguishes things that should not be distinguished.

2. Which is it?

Problem	Which	Count
President, VP, secretary from 12	Permutation	$P(12,3) = 1320$
Three-person committee from 12	Combination	$C(12,3) = 220$
Five-card hand from 52	Combination	$C(52,5) = 2{,}598{,}960$
Bit strings of length 10 with exactly four 1s	Combination	$C(10,4) = 210$

The bit-string row is the one to study

Choosing a bit string with four 1s means choosing *which four of the ten positions* hold the 1s — an unordered choice of positions. Many problems that do not look like selections are combinations once you ask *what is being chosen*.

3. Two identities

Theorem 8.46

For $0 \le r \le n$: $\binom{n}{r} = \binom{n}{n-r}$.

Why, without algebra. Choosing r elements to *include* is the same as choosing the $n-r$ to leave out. The two choices determine each other.

So $C(10,4) = C(10,6) = 210$, and you may always compute the easier one.

The subset identity

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Why. The left side counts subsets by their size; the right side counts all subsets at once, by deciding in-or-out for each element. Both count the same thing, so they are equal.

That is Theorem 2.17 from Lecture 11, arrived at a second way. **Counting one collection two ways and equating the answers** is how most combinatorial identities are proved.

4. Combining with the other principles

A committee of 5 from 7 women and 6 men, with exactly 3 women.

Choose the women *and then* the men — Multiplication Principle:

$$C(7,3) \cdot C(6,2) = 35 \cdot 15 = 525.$$

Bit strings of length 10 with at least eight 1s.

Eight, nine, or ten 1s — disjoint cases, Addition Principle:

$$C(10,8) + C(10,9) + C(10,10) = 45 + 10 + 1 = 56.$$

Multiply within, add across

Independent stages of one selection multiply. Mutually exclusive cases add. Most problems from here on use both, and deciding which applies where is the whole task.

Exit ticket

1. Compute $C(9,4)$ and $C(9,5)$. Why are they equal?

2. How many 4-person committees from 10 people include a specific person?

Before Friday

Read Section 9.1, "The Pascal Triangle and the Binomial Theorem," pp. 313–322.

Self-check (not collected). Section 8.4, Exercises 7, 9, and 11 — odd-numbered, answers in the back.

Homework 12 is due Friday, November 13, 11:59 p.m.

Questions I still have